EECE 306 Lab 8 notebook. Dielectrics, Boundary Conditions, and Capacitance

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This notebook collects the Lab 8 calculations, checks, and design curve. We used harmonic face permittivity for the solver and discrete face gradients for the stored energy, so both calculations use the same grid.

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Setup

```
clear; close all;
addpath(pwd);
e0 = em.const.eps0();
run(fullfile('tests','lab08_studies.m'));
```

Rung 1. Parallel plate against the formula

Plates span the full width, so the field between them is exactly one dimensional and C per unit depth must equal eps0 times w over d.

```
n = 82;
G = em.solve.grid2d([0 1], [0 0.1], n, n);
iy1 = round(0.25*(n-1)) + 1;  iy2 = round(0.75*(n-1)) + 1;
mask = false(n,n); mask(iy1,:) = true; mask(iy2,:) = true;
Vfix = zeros(n,n); Vfix(iy2,:) = 1;
C = em.derive.capacitance(G, 1, mask, Vfix, ...
    struct('method','direct'));
d = G.y(iy2)-G.y(iy1);
Cref = e0 / d;
fprintf('parallel plate C %.5e; formula %.5e; error %.2e\n', ...
    C, Cref, abs(C-Cref)/Cref);
```

```
parallel plate C 1.74924e-10; formula 1.74924e-10; error 5.91e-15
```

Two dielectric layers in series

```
epsr = ones(n,n); epsr(G.Y < 0.05) = 4;
C2 = em.derive.capacitance(G, epsr, mask, Vfix, ...
    struct('method','direct'));
C2ref = e0 / ((d/2)/4 + (d/2)/1);
fprintf('two layers C %.5e; formula %.5e; error %.2e\n', ...
    C2, C2ref, abs(C2-C2ref)/C2ref);

% The plate gap is about 0.051 times the width. The side walls are insulating,
% so this is an ideal one-dimensional check with no edge fringing. It does
% not prove that a finite, open plate capacitor has negligible fringing.
```

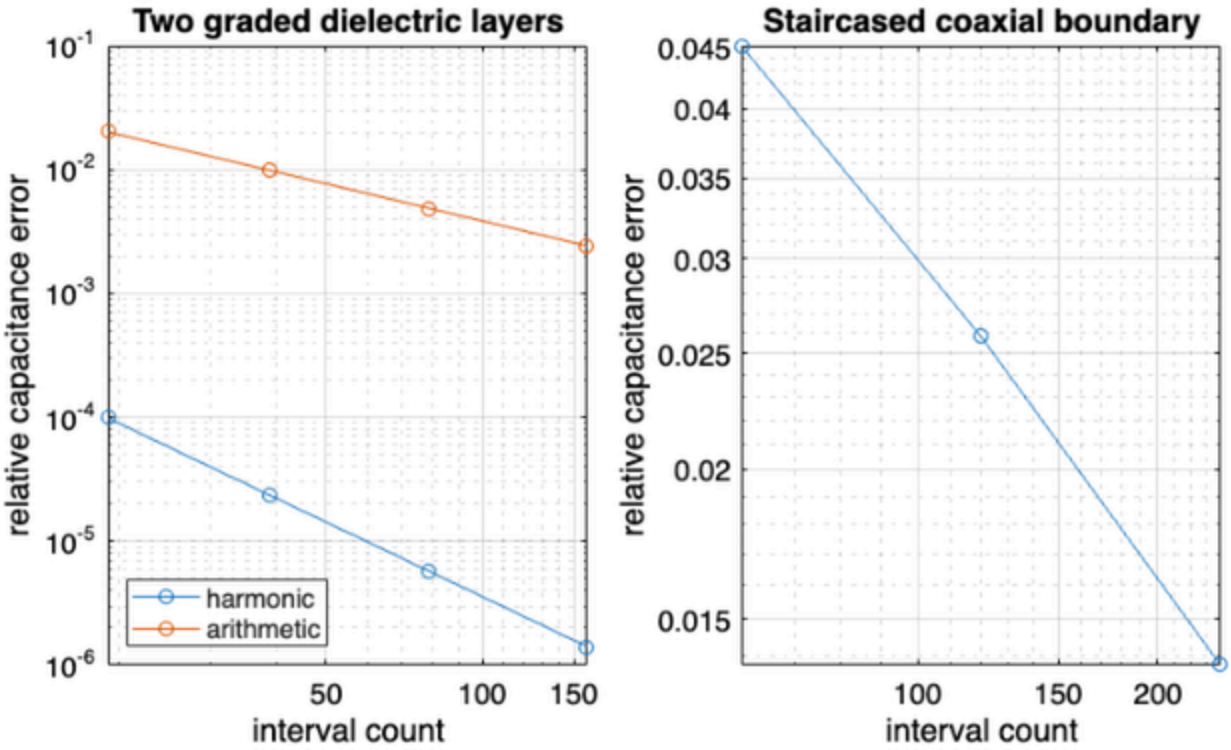
```
two layers C 2.79879e-10; formula 2.79879e-10; error 6.83e-15
```

Interface averaging and grid refinement

The constant layers match the series result to round-off when the jump falls between nodes. There is no useful convergence slope left to fit. For the order check, each layer varies as (1+y), with a factor-of-four jump at y=0.5. The exact series integral is log(1.5)/4+log(2/1.5). This averaging comparison measures terminal face flux. The capacitance API itself uses stored energy with harmonic faces.

```
disp('N      harmonic relative error      arithmetic relative error')
fprintf('%4d       %12.4e              %12.4e\n',[S.N S.interfaceErrors]');
fprintf('graded-layer order: harmonic %.3f, arithmetic %.3f\n', S.pH, S.pA);
figure;
subplot(1,2,1); loglog(S.N-1,S.interfaceErrors,'o-'); grid on
xlabel('interval count'); ylabel('relative capacitance error')
legend('harmonic','arithmetic','Location','southwest')
title('Two graded dielectric layers')
subplot(1,2,2); loglog(S.coaxN-1,S.coaxErrors,'o-'); grid on
xlabel('interval count'); ylabel('relative capacitance error')
title('Staircased coaxial boundary')
```

```
N      harmonic relative error      arithmetic relative error
  20         9.8910e-05               2.0107e-02
  40         2.3472e-05               9.8445e-03
  80         5.7202e-06               4.8706e-03
 160         1.4121e-06               2.4225e-03
graded-layer order: harmonic 2.000, arithmetic 0.996
```



Rung 2. Coaxial line against the log formula

```
n = 241;
Gc = em.solve.grid2d([-1 1], [-1 1], n, n);
R = sqrt(Gc.X.^2 + Gc.Y.^2);
a = 0.3; b = 0.9;
maskC = R <= a | R >= b;
VfixC = zeros(n,n); VfixC(R <= a) = 1;
Cc = em.derive.capacitance(Gc, 1, maskC, VfixC, ...
    struct('method','direct'));
```

```
CcRef = 2*pi*e0 / log(b/a);
fprintf('coax C %.5e; formula %.5e; error %.2e\n', ...
    Cc, CcRef, abs(Cc-CcRef)/CcRef);
disp('coax N      capacitance (F/m)      relative error')
fprintf('%4d       %12.5e           %12.4e\n',[S.coaxN S.coax S.coaxErrors]');
fprintf('coax observed order %.3f\n',S.pCoax);
% The coax capacitance comes up toward the formula as the grid gets finer.
% The circular boundaries are staircased, so the measured order is closer
% to one than two even though the interior stencil is second order.
```

```
coax C 4.99435e-11; formula 5.06389e-11; error 1.37e-02
coax N      capacitance (F/m)      relative error
  61        4.83529e-11             4.5143e-02
 121        4.93311e-11             2.5825e-02
 241        4.99435e-11             1.3731e-02
coax observed order 0.859
```

The two blunder catchers

```
VfixC100 = zeros(n,n); VfixC100(R <= a) = 100;
Cc100 = em.derive.capacitance(Gc, 1, maskC, VfixC100, ...
    struct('method','direct'));
fprintf('C at 1 V: %.5e; C at 100 V: %.5e\n', Cc, Cc100);
% A missing voltage square would make these differ by 1e4.
Cfill = em.derive.capacitance(Gc, 3, maskC, VfixC, ...
    struct('method','direct'));
fprintf('epsr=3: C_filled/C = %.4f (expect 3)\n', Cfill/Cc);
```

```
C at 1 V: 4.99435e-11; C at 100 V: 4.99435e-11
epsr=3: C_filled/C = 3.0000 (expect 3)
```

Resistance of a uniform bar

```
n = 81;
Gr = em.solve.grid2d([0 1], [0 1], n, n);
maskR = false(n,n); maskR(:,1) = true; maskR(:,end) = true;
VfixR = zeros(n,n); VfixR(:,1) = 1;
sigma = 100;
Rbar = em.derive.resistance(Gr, sigma, maskR, VfixR, ...
    struct('method','direct'));
Rref = 1 / (sigma * 1);
fprintf('bar R %.5e ohm m; L/(sigma*w) %.5e ohm m\n', Rbar, Rref);
```

```
bar R 1.00000e-02 ohm m; L/(sigma*w) 1.00000e-02 ohm m
```

Rung 3. The payoff. A microstrip design curve

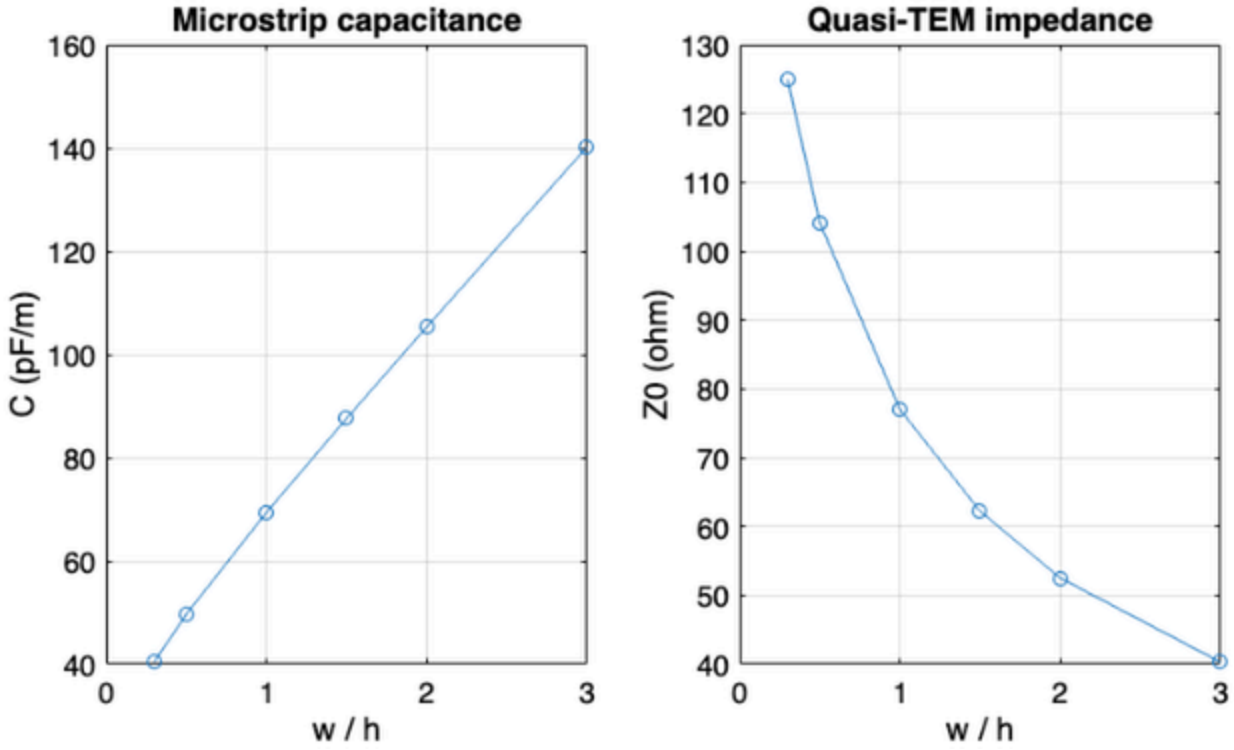
The strip is h=0.2 m above ground, with epsr=4 below it and air above. The grid covers x=[-2,2], y=[0,2] m at 0.01 m spacing. The outer walls are grounded, so this is a boxed approximation to an open microstrip. We solve the same geometry with air to get C0, then use Z0 = 1/(c0*sqrt(C*C0)) for the lossless, nonmagnetic quasi-TEM model.

```
widths = S.widths; hsub = S.hsub; Cw = S.Cw;
figure;
subplot(1,2,1); plot(widths/hsub,Cw*1e12,'o-'); grid on
xlabel('w / h'); ylabel('C (pF/m)'); title('Microstrip capacitance')
subplot(1,2,2); plot(widths/hsub,S.Z0,'o-'); grid on
xlabel('w / h'); ylabel('Z0 (ohm)'); title('Quasi-TEM impedance')
disp('w/h       C (pF/m)       C0 air (pF/m)       Z0 (ohm)')
disp([widths(:)/hsub Cw(:)*1e12 S.Cair(:)*1e12 S.Z0(:)])
fprintf('widest strip: C %.3f pF/m; plate estimate %.3f pF/m\n', ...
    Cw(end)*1e12,S.ppWide*1e12);
fprintf('widest strip: mesh change %.2f%%; larger box change %.2f%%\n', ...
    100*S.meshChange,100*S.boxChange);
```

```
% Making the strip wider raises C and lowers Z0 over the decade we checked.
% At w/h=3, C is about 140.2 pF/m versus 106.3 pF/m from the plate estimate;
% the extra capacitance comes from fringing, with some influence from the box.
% Halving the grid spacing still changes the widest case by about 6.7%, so
% these are design estimates, not fully mesh-converged open-line values.
% Enlarging the box at fixed coarse spacing changes that case by about 0.22%.
```

```
w/h       C (pF/m)       C0 air (pF/m)       Z0 (ohm)
    0.3000   40.4600   17.6213  124.9244
    0.5000   49.6827   20.6406  104.1635
    1.0000   69.4020   27.0283   77.0166
    1.5000   87.6156   32.7926   62.2301
    2.0000  105.3425   38.2909   52.5206
    3.0000  140.2139   48.8872   40.2890

widest strip: C 140.214 pF/m; plate estimate 106.250 pF/m
widest strip: mesh change 6.70%; larger box change 0.22%
```



Boundary conditions and what the numbers mean

```
fprintf('tangential E jump %.3e V/m; normal D relative spread %.3e\n', ...
    S.tangentError,S.normalDspread);
fprintf('uniform geometry RC %.6e s; eps/sigma %.6e s\n',S.RC,S.RCref);
% The tangential-field check uses a nonzero field on both sides of the
% interface, so a zero-field symmetry case cannot make it pass by accident.
% Normal D stays continuous without free interface charge. Conductivity
% replaces permittivity in the steady-current equation; for one uniform
% geometry, C per depth times R times depth is eps/sigma, in seconds.
% The energy sums harmonic face coefficients times squared voltage drops.
% Resistance instead sums current over faces leaving the high terminal.
% Geometry and interface placement still limit the accuracy, so a passing
% formula check does not make the boxed microstrip an exact open-line result.
```

```
tangential E jump 3.109e-14 V/m; normal D relative spread 1.933e-13
uniform geometry RC 2.656256e-13 s; eps/sigma 2.656256e-13 s
```

Full regression suite, Lab 1 through Lab 8

```
runTests
```

```
EECE 306 toolbox test suite
------------------------------------------------------------

eta =

  376.7303

  test_lab01 checks complete
  PASS  test_lab01            0.24 s
  test_lab02 checks complete
  PASS  test_lab02            0.12 s
  5000-point field evaluation: 0.0033 s
  test_lab03 checks complete
  PASS  test_lab03            0.10 s
  test_lab04 checks complete
  PASS  test_lab04            0.20 s
  merged source: flux 5.00000000e-09 C, enclosed 5.00000000e-09 C
  density errors at 128000 / 1024000 elements: 0.953% / 1.715% max
  sphere / cube / closed hemisphere: 3.00000000e-09 / 3.00000000e-09 / 3.00000000e-09 C
  sphere Gauss rel err 1.42e-11; midpoint N=960 rel err 4.46e-07
  test_lab05 checks complete
  PASS  test_lab05            1.28 s
  Lab 6: max absolute map error 7.776e-04 V/m
  max closed residual / (max|E| L) 2.473e-13
  Square vs exact-tangent edges 4.178e-11 V
  open midpoint errors [0.0004329 0.0001082 2.706e-05 6.764e-06] V (order 2)
  Fixed line refinement 4.965e-06 V
  growing line potentials 25.9495 -> 37.6527 V
  warning verified
  Step minimum h=3.162e-06 m, error 1.025e-11
  default error 3.586e-11
  10000 points V/grad 0.001/0.008 s
  test_lab06 checks complete
  PASS  test_lab06            1.69 s
  Lab 7: MMS error 8.224e-05, order 2.042
  Jacobi / GS / SOR iterations 1320 / 689 / 77
  Gauss flux 2.5947e-09 C, enclosed 2.5000e-09 C
  test_lab07 checks complete
  PASS  test_lab07            0.26 s
  Lab 8: plate error 1.110e-16; constant-layer error 4.441e-16
  Graded-layer orders harmonic 2.000; arithmetic 0.996
  Coax finest error 1.373%; order 0.859
  Voltage ratio 1.0000000000; filled ratio 3.0000000000
  Bar R 0.01000000 ohm m; tangential E jump 3.109e-14 V/m
  test_lab08 checks complete
  PASS  test_lab08            1.13 s
------------------------------------------------------------
  8 passed, 0 failed, 8 total
  All tests passing.
```

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